

## Multiple Choice (10 marks)

1. There are 230 calories in 4 ounces of a type of ice cream. How many calories are in 6 ounces of that ice cream?
  - (a) 232
  - (b) 236
  - (c) 345
  - (d) 460
2. A zoo has 15 toucans and 60 parrots. What is the ratio of the number of toucans to the number of parrots at the zoo?
  - (a) 1:04
  - (b) 1:05
  - (c) 4:01
  - (d) 4:05
3. Balloons come in packages of 8. If you need 42 balloons, how many packages should you buy?
  - (a) 5
  - (b) 6
  - (c) 8
  - (d) 42
4. Suppose two events, E and F, have nonzero probabilities  $p$  and  $q$ , respectively. Which of the following is impossible?
  - (a)  $p + q > 1$
  - (b)  $p - q < 0$
  - (c)  $p / q > 1$
  - (d) E and F are both independent and mutually exclusive.
5. Let  $p$ ,  $q$ , and  $r$  be constants. One solution to the equation  $(x - p)(x - q) = (r - p)(r - q)$  is  $x = r$ . Find the other solution in terms of  $p$ ,  $q$ , and  $r$ .
  - (a)  $pqr$
  - (b)  $pq-r$
  - (c)  $p-q-r$
  - (d)  $p+q-r$

## True / False (10 marks)

*Answer True or False*

6. True or False: To find the number of students in each of Ms. Carter's 5 groups, divide 30 by 5.
7. True or False: The correct answer to 'Given that  $n > 1$ , what is the smallest positive integer  $n$  whose positive divisors have a product of  $n^6$ ?' is '64'.
8. True or False: The greatest common factor of 252 and 96 is 12.
9. True or False: Randomization in assigning subjects to different treatments is unimportant in the design of experiments.
10. True or False: The five-digit number 24,516 has only 5 integers from 1 to 9 that are its divisors.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Determine the day of the week that Carlos Montado will be 706 days old, beginning from his birth on November 9, 2002. Explain your reasoning and calculations in detail.
12. Dawn needs to repair her windows by purchasing materials. Calculate the total cost of 3 feet of wood and 4 pieces of glass. Explain your reasoning and show all steps in your calculation.

*End of Paper*